Observational Indistinguishability and Integrity Blind Regions in Hybrid Quantum-Classical Workflows

Roberto Fernández-Barrios, Iker Pastor-López, Amaia Pikatza-Huerga, and Pablo García Bringas

Abstract—We model hybrid quantum-classical integrity through observational indistinguishability under declared views and references. The result is claim-relative minimum evidence granularity within the declared reference lattice: a trusted same-batch scalar claim reference, a trusted aggregate and trusted item-aligned evidence certify conclusion, aggregate and item-identity claims, respectively; an external trust root remains necessary. Across eight frozen CICIDS2017, UNSW-NB15 and ToN-IoT environments, 3,600 label interventions confirm the structural boundaries. Structural blindness is distinct from finite-sample statistical miss and independent of batch size; executed detection is not a universal power bound. The equal-weight grid is a design summary, not deployment prevalence. The design calibrates on clean resamples; a geometry-aligned sensitivity fixes the reference/batch construction between clean and intervention. Mean shift and scaling remain almost fully responsive, dropout strength-dependent, and a adaptive cluster-preserving intervention reduces response against matched controls in 25–40 of 40 environment/split cells while retaining materially altered audit results. A full-conformal Max-Rank rule is valid under exchangeability; executed false-action rates are descriptive. A bounded 165-design-cell ideal-statevector and finite-shot-emulation gate instantiates the lattice without a hardware claim.

Index Terms—hybrid quantum-classical systems, integrity auditing, observability, quantum kernels, workflow integrity

1 Introduction

Hybrid quantum-classical software distributes one reported result across heterogeneous computational and evidential boundaries. Classical records are acquired, cleaned, projected and encoded; a circuit and execution stack produce quantum evidence; a learner turns a kernel into a decision; and an evaluation layer aligns that decision with reference outcomes and reports a metric. A plausible final number may therefore coexist with a failure in the data, circuit, kernel, prediction, label or reporting path. Security analyses of hybrid systems consequently call for lifecycle-aware controls rather than trust in an isolated algorithm [1], [2].

- *The authors are with the Faculty of Engineering, University of Deusto, Avda. de las Universidades 24, 48007 Bilbao, Spain. E-mail: roberto.fernandez.b@deusto.es (corresponding author), iker.pastor@deusto.es, a.pikatza@deusto.es, pablo.garcia.bringas@deusto.es.*
- *This work is part of grant PID2024-155693NB-C43, ATHENA-AEGIS (Advanced Secure Technologies for Hybrid Quantum-Classical Environments and Applications), funded by MICIU/AEI/10.13039/501100011033 and by ERDF/EU.*

Robustness benchmarking collapses this chain into a performance change. That endpoint cannot identify what changed or whether the auditor could have seen it. Consider a hybrid security service that scores each batch with a fixed predictor, then joins the predictions to a ground-truth feed from another subsystem (analyst verdicts, incident tickets or a delayed label store). The predictor is intact, but the join or store is stale, corrupted or rewritten, so balanced accuracy moves although the model has not. Feature and prediction monitors cannot see the event because their inputs are unchanged. Label counts expose an unbalanced corruption but not a balanced exchange of class identities. A signature authenticates the label file, not its item-level alignment with the predictions, so a correctly signed stale or misaligned file still passes.

Calling the event stealthy therefore requires a qualifier: stealthy to which auditor, holding which evidence and trusting which reference? The constructive result is claim-relative. A trusted scalar reference R_0 for the same batch certifies whether the reported conclusion changed. A trusted aggregate M_0 additionally certifies aggregate integrity and, by Proposition 3, every material conclusion change in the declared setting; only item-identity integrity requires item alignment. If one party controls both the label store and its checking reference, no local verifier can distinguish a substituted honest pair from the original (Proposition 6). The paper identifies the minimum external root and its granularity within the declared reference model for the protected claim; it does not remove the need for that root. The scenario is a reading aid, not an incidence estimate.

Prior art already establishes observation-relative detectability [3]–[5], authenticates evaluation artifacts through VAMP [6], constructs drifts that evade their detectors [7], and verifies quantum-stage contracts and behavioural evidence on hardware [8], [9]. We claim none of these ingredients. The novelty lies in their composition as a claim-relative evidence lattice for hybrid workflow integrity: it derives structural blind regions and the information/reference granularity sufficient or insufficient for conclusion, aggregate and item-identity claims, explicitly including the evaluation-label path and separating structural blindness from finite-sample statistical miss. The experiments validate its structural predictions, measure conformal-rule policy consequences, instantiate residual adaptive evasion against a declared fingerprint, and map the distinctions onto a bounded quantum branch.

The contribution is one package:

- a claim-relative workflow view/reference lattice, with monotonicity, closure, materiality and minimal counterexamples, including distinct semantic, estimated and observed kernels;
- an explicit adversary/failure model, including an adaptive attacker that preserves the declared batch fingerprint;
- multi-dataset fixed-OOD validation with verified deduplication, cluster-level uncertainty and prespecified null calibration;
- a conformal family rule valid under exchangeability and an offline `allow/hold/block` analysis over the frozen grid; and
- a bounded quantum-path instantiation and four-contract prototype.

The evidence lattice and its claim-relative derivation are the contribution; the conformal Max-Rank construction, contracts, hash chains, cryptographic roots, unobservable attacks and semantics-preserving tests remain prior-art building blocks. The secondary quantum-versus-classical impact profile (supplement) is neither a quantum-advantage result nor a vulnerability ordering. Physical hardware, calibrated backend noise, scheduling, provider-side security, multi-tenant interference, deployed runtime services and operational Fleet Management are outside the claim.

2 Related Work and Positioning

2.1 Stealth, Detectability and Observability in Secure Systems

Observation-relative detectability is established: cyber-physical attacks can be unobservable, residual-stealthy or confined to an estimator's range space [3]–[5]; later work quantifies the detectability–impact trade-off and the residuals a monitor can form [10]–[12]. Hinder *et al.* construct drifts designed to evade drift detectors [7]. Gate A is a concrete cluster-preserving instance against our declared batch fingerprint; it measures materiality, policy consequences and evidence regimes without claiming priority for adaptive drift evasion. Simplex supplies the canonical decision/fallback pattern [13], [14], runtime verification checks executions [15], and integrity checkers compare digests with trusted baselines [16]. Our claim is not generic information-relative detectability, but blind regions and calibrated decisions for workflow views whose protected boundaries include evaluation ground truth.

2.2 Monitoring, Labels and Evaluation Integrity

Monitoring work treats shifted priors, selective labels and proxy sufficiency under delayed ground truth [17]–[20]. Benchmark-quality work treats label error, evaluation pitfalls, spatial/temporal sampling bias and measurement blindness [21]–[24]. These lines primarily model labels as shifted, delayed, noisy, erroneous or measurement-limited. VAMP [6], by contrast, explicitly treats evaluation artifacts as authenticated assets that may be poisoned or substituted. Our distinction is therefore not that evaluation data can be attacked, but that auditability and trusted-reference granularity are derived relative to the protected claim and the auditor's view.

VAMP protects named datasets, software, models and evaluation sets through authenticated provenance, providing a concrete external root of the kind required by Proposition 6. Its focus is artifact authentication; our complementary question is which view and granularity suffice for conclusion, aggregate or item-identity integrity, and what remains indistinguishable with weaker evidence. A VAMP-style manifest can instantiate the references assumed here.

In-toto, SLSA and Sigstore bind supply-chain steps, build provenance, identity and transparency [25]–[27]. Remote attestation supplies fresh evidence to an appraiser [28]; RATS names Attesters, Verifiers, Evidence, Reference Values and appraisal policy [29]. These mechanisms establish provenance or appraisal, not whether aggregate or item-aligned evidence suffices for a claim. Provider/runtime authentication belongs to separate operational work.

2.3 Calibration of Multi-Sensor Decisions

The family rule of Section III-D is a conformal p-value [30], [31]. We use the Max-Rank statistic of Timans et al. within a full-conformal family construction [32]: the maximum of the per-sensor ranks is equivalently a Tippett minimum-p ordering, closely related to Westfall–Young resampling corrections [33], [34]. The augmented leave-one-out construction and ties-against-rejection convention supply the stated finite-sample argument under exchangeability [30], [35]. Conformal anomaly detection and shared-calibration p-values provide the surrounding statistical context [36], [37]. We claim no novelty for the statistic or rule: the family score calibrates an information-regime decision under its stated premise and exposes the executed design's premise violation. Beyond-exchangeability conformal methods exist [38], but this paper does not implement one. The asymmetric comparison rule had no such guarantee (Proposition 5(c)).

2.4 Hybrid and Quantum Software Integrity

Hybrid-system analyses decompose control, compilation, execution and measurement surfaces and cover malicious compilation, hardware faults, cloud exposure and side channels [1], [2], [39]. Quantum Vulnerability Factor scores fault sensitivity, while Quantum Leak demonstrates cloud timing leakage [40], [41]. QCIVET combines stage contracts, hash-chained traces, behavioural subtyping, calibrated observable tests and QPU validation [8]. QML-PipeGuard uses a structured observable family and calibrated drift tolerance to detect channel substitution on IBM hardware [9]; multi-level work evaluates circuit-integrity metrics [42].

Bensoussan *et al.* derive a taxonomy from real hybrid-software faults [43]; our synthetic interventions are controlled distinguishability probes, not a prevalence model. Quantum Squeeziness formalizes quantum-program testability and fault masking [44]; we instead study workflow-wide observational indistinguishability across a claim-relative evidence lattice, including evaluation labels and trusted-reference granularity.

QProv records provider-independent provenance across circuit, compilation, execution and hardware context [45]; recent cross-provider work extends this line toward unified quantum-job provenance [46]. Differential, metamorphic and equivalence-modulo-input tests address the oracle problem [47]–[49]; compiler verification, program logics and runtime assertions supply other formal/runtime mechanisms [50]–[54]. We claim no priority for these mechanisms. QCIVET and QML-PipeGuard reason directly about quantum-stage contracts and behavioural evidence, including calibrated observables, drift and hardware validation; QProv and cross-provider work address provenance. They are stronger here on QPU/hardware and runtime/provider evidence. Our orthogonal contribution is a workflow-wide, claim-relative information-set lattice that protects the evaluation-label path and distinguishes conclusion, aggregate and item-identity integrity. Its quantum evidence is simulator-only, its hash chain unsigned, and its policy offline.

2.5 Relation to Companion Work

Three companion studies by the authors are disclosed to delimit the claim. *Sharp target-domain certificates* [55] bounds the advantage of a fixed quantum-kernel candidate under distribution shift (shared data sources and fidelity kernels; an identification interval, not the observability of interventions). *Conditional validity of quantum event classifiers* [56], the closest relative, conditions fail-closed verdicts on a declared information set and exhibits exact sensor blindness to weight-only nuisances in collider data; here the conditioning is applied to the integrity of the evidence chain under interventions, with trusted references, a conformal-rule policy and executable response. *Candidate comparability before promotion* [57] decides whether a challenger should replace a deployed detector after a drift alarm (shared benchmarks and label-free monitors). The present study concerns workflow integrity, observational indistinguishability, evidence granularity and integrity policy. A separate longitudinal study concerns operational QPU and context assurance. Data sources, fidelity-kernel implementations and some monitors are shared and are not presented as independent contributions; none of the propositions, interventions, tables or experiments reported here appears in the three companion manuscripts.

3 Observation Model and Blind Regions

The supplement gives complete proofs. The formal core derives which evidence separates each intervention class and integrity claim.

3.1 States, Interventions and Materiality

A workflow state separates stored artifacts from those the workflow derives. The primitive artifacts of one evaluation run are

$$s = (X, P, C, E, \xi, g, f, y) \in \mathcal{S}, \tag{1}$$

acquired features X, preprocessing P, circuit or feature map C, execution context E, estimation randomness ξ, post-processing map g, fitted predictor f and evaluation labels y. The derived artifacts are recomputed from them: the representation $\tilde{X} = P(X)$; the semantic kernel $K_{\text{sem}} = \Phi(C)$ induced by the circuit on a fixed probe set; the finite-shot estimate $\hat{K} = \text{est}(K_{\text{sem}}, E, \xi)$; the observed kernel $K_{\text{obs}} = g(\hat{K})$ delivered downstream after estimation, transmission and post-processing; the predictions $\hat{y} = f(\tilde{X}, K_{\text{obs}})$, indexed by item identity $i = 1, \ldots, n$; and the result $R = r(\hat{y}, y)$, balanced accuracy of $\hat{y}$ against y. An intervention a overwrites a declared set of artifacts, primitive or derived, keeps every non-descendant fixed and recomputes the descendants through the workflow with ξ retained; the baseline is s_0 and $s_a = a(s_0)$. Label-only T_y overwrites y item-wise, so only R is recomputed (its prior-preserving subclass T_y^{π} also preserves the class histogram); feature-side mechanisms overwrite $\tilde{X}$ or X and recompute $\hat{y}$ and R; circuit-side interventions overwrite C and recompute K_{sem} and everything below it; estimation variation overwrites ξ or E with K_{sem} fixed; post-processing overwrites g or K_{obs} with K_{sem} and $\hat{K}$ fixed. “Fixes everything else” thus means every non-descendant. The signed conclusion change is

$$\Delta_R(a) = R(s_0) - R(s_a). \tag{2}$$

An intervention is *material at level* τ if $|\Delta_R(a)| \geq \tau$. The limit $\tau \to 0^+$ is a structural-sensitivity endpoint—whether the reported conclusion changes at all—not an operational risk threshold or an assertion that every epsilon has the same consequence. The fixed $\tau = 0.02$ and 0.05 analyses show larger effect sizes; service-level risk is outside this study. Earlier versions used the positive part $\max(\Delta_R, 0)$; the signed endpoint counts an apparent improvement caused by an integrity failure as a conclusion change.

3.2 Views, Refinement and Trusted References

An information set $\mathcal{I}$ is a view map $V_{\mathcal{I}} : \mathcal{S} \to \mathcal{O}_{\mathcal{I}}$; the auditor observes $V_{\mathcal{I}}(s)$ and nothing else. The regimes studied are $\mathcal{I}_X$ (the multiset of rows of $\tilde{X}$), $\mathcal{I}_{XF}$ (the multiset of pairs $(\tilde{x}_i, \hat{y}_i)$), $\mathcal{I}_{Y_m}$ (the class-count vector of y), $\mathcal{I}_{XFY}$ (the multiset of triples $(\tilde{x}_i, \hat{y}_i, y_i)$) and $\mathcal{I}_Q$ (circuit, kernel and execution evidence). $\mathcal{I} \sqsubseteq \mathcal{I}'$ (refinement) iff $V_{\mathcal{I}} = \pi \circ V_{\mathcal{I}'}$ for some π; hence $\mathcal{I}_X \sqsubseteq \mathcal{I}_{XF} \sqsubseteq \mathcal{I}_{XFY}$ and $\mathcal{I}_{Y_m} \sqsubseteq \mathcal{I}_{XFY}$, while $\mathcal{I}_{Y_m}$ is incomparable with $\mathcal{I}_X$ and $\mathcal{I}_Q$ is a separate branch. The join $\mathcal{I} \vee \mathcal{W}$ is the pair of views.

A *reference profile* separates three dimensions: provenance (historical, benchmark-protected, or deployment-authenticated), granularity (aggregate or item-aligned), and decision rule (statistical or exact). A reference is *trusted* if no intervention in the class under study can alter it. The A–C labels are shorthand, not a one-dimensional trust scale. Class A is a *statistically thresholded aggregate comparison*; its reference may be historical or, as executed here, a protected clean same-item-set oracle used without item pairing. The oracle is outside the benchmark attack API, but no deployed authentication mechanism is demonstrated. Class B is an exact invariant against a trusted aggregate same-batch reference, such as $M(s_0)$, $\text{hist}(y_0)$ or $R(s_0)$. Class C is an exact invariant against a trusted item-aligned same-batch reference such as $(y_{0,i})_i$. Thus statistical versus exact and aggregate versus item-aligned are not synonyms for untrusted versus trusted. For a label-path intervention, *item-identity integrity* holds if $y_a = y_0$ item-wise, *aggregate integrity* if $M(s_a) = M(s_0)$ and *conclusion integrity* if $R(s_a) = R(s_0)$; each implies the next and no converse holds. $\mathcal{I}^\star$ denotes a regime augmented with trusted same-batch references ($\mathcal{I}^\star_{XFY}$ holds the aggregate confusion profile and item-aligned labels and predictions; feature coverage remains statistical). A label available in $\mathcal{I}_{XFY}$ is not thereby trusted:

if the label store is the asset under attack, the auditor sees y_a, not y_0.

3.3 Equivalence, Sensors and Blind Regions

Definition 1. $s \sim_{\mathcal{I}} s'$ iff $V_{\mathcal{I}}(s) = V_{\mathcal{I}}(s')$. A sensor admissible in $\mathcal{I}$ is a measurable $S : \mathcal{O}_{\mathcal{I}} \to \mathbb{R}^k$, written $S(s) = S(V_{\mathcal{I}}(s))$.

Lemma 1 (structural blindness). If $s_a \sim_{\mathcal{I}} s_0$ then $S(s_a) = S(s_0)$ for every sensor admissible in $\mathcal{I}$ (path-wise for a seeded randomized sensor; in distribution for a fresh seed).

Definition 2. For a class $\mathcal{A}$, the structural blind region of $\mathcal{I}$ is $B_{\mathcal{I}}(\mathcal{A}) = \{a \in \mathcal{A} : a(s_0) \sim_{\mathcal{I}} s_0\}$ and the sensor blind region of a family $\mathcal{S}$ is $B_{\mathcal{I},\mathcal{S}}(\mathcal{A}) = \{a : S(a(s_0)) = S(s_0)\ \forall S \in \mathcal{S}\}$. By Lemma 1, $B_{\mathcal{I}} \subseteq B_{\mathcal{I},\mathcal{S}}$, with equality iff $\mathcal{S}$ is *baseline-separating* on the orbit of $\mathcal{A}$ (a different value at $a(s_0)$ than at s_0 whenever the views differ); detection needs baseline separation, identification would need pairwise separation, which is not claimed. Sensor blindness thus decomposes into information-level blindness and sensor insufficiency; the adaptive attacker of Section VI-D exploits the second. The structural and sensor auditability gaps at level τ are

$$G_{\mathcal{I}}(\tau) = \{a : |\Delta_R(a)| \geq \tau,\ a \in B_{\mathcal{I}}\}, \quad G_{\mathcal{I},\mathcal{S}}(\tau) \supseteq G_{\mathcal{I}}(\tau). \tag{3}$$

Proposition 1 (monotonicity). If $\mathcal{I} \sqsubseteq \mathcal{I}'$ then $B_{\mathcal{I}'}(\mathcal{A}) \subseteq B_{\mathcal{I}}(\mathcal{A})$ and $G_{\mathcal{I}'}(\tau) \subseteq G_{\mathcal{I}}(\tau)$ for every τ.

Proof: $V_{\mathcal{I}'}(s_a) = V_{\mathcal{I}'}(s_0)$ implies $V_{\mathcal{I}}(s_a) = \pi(V_{\mathcal{I}'}(s_a)) = \pi(V_{\mathcal{I}'}(s_0)) = V_{\mathcal{I}}(s_0)$. □

Corollary 1 (label-path boundaries). Let f be fixed and deterministic. (a) $T_y \subseteq B_{\mathcal{I}_{XF}} \subseteq B_{\mathcal{I}_X}$: every sensor admissible in $\mathcal{I}_X$ or $\mathcal{I}_{XF}$ is invariant under a label-only change. (b) $T_y^{\pi} \subseteq B_{\mathcal{I}_{Y_m}}$: every sensor admissible in $\mathcal{I}_{Y_m}$ is invariant under a prior-preserving change.

Proposition 2 (closure). $B_{\mathcal{I} \vee \mathcal{W}}(\mathcal{A}) = B_{\mathcal{I}}(\mathcal{A}) \cap B_{\mathcal{W}}(\mathcal{A})$; a class $\mathcal{A} \subseteq B_{\mathcal{I}}$ becomes completely separable after adding $\mathcal{W}$ iff $V_{\mathcal{W}}(a(s_0)) \neq V_{\mathcal{W}}(s_0)$ for every $a \in \mathcal{A}$.

Corollary 2 (label-path closure). No view factoring through $(\tilde{X}, \hat{y}, \mathrm{hist}(y))$ separates any $a \in T_y^{\pi}$. The multiset $\mathcal{I}_{XFY}$ separates it unless labels move only among items with identical $(\tilde{x}, \hat{y})$; an item-aligned view separates every non-identity relabeling.

Proposition 3 (materiality forces aggregate separability). Let $R = g(M)$ be a function of the confusion matrix $M(s)$, as balanced accuracy is. If $a \in T_y$ and $\Delta_R(a) \neq 0$ then $M(s_a) \neq M(s_0)$; hence a is separable in the confusion view and in $\mathcal{I}_{XFY}$, and $G_{\mathcal{I}_{XFY}}(\tau) \cap T_y = \emptyset$ for every $\tau > 0$.

Proof: Contrapositive of $M(s_a) = M(s_0) \Rightarrow R(s_a) = R(s_0)$; the confusion view is a coarsening of the multiset of triples. □

The proposition holds for every metric that is a function of the binarised confusion matrix; for a ranking metric such as ROC-AUC the same argument holds with the multiset of (score, label) pairs as the aggregate (supplement). No AUC experiment is run.

3.4 Three Auditor Classes and Decision-Level Calibration

A *reference-anchored* auditor holds a trusted same-batch reference ρ (class B or C) and computes $D(s) = d(V_{\mathcal{J}}(s), \rho)$ with a metric d; then $D(s_0) = 0$ exactly, the rule "fire iff $D > 0$" has zero false-action probability, and separability in $\mathcal{J}$ is sufficient for detection. A *batch-level statistical* auditor computes aggregate scores and thresholds them against a clean calibration population (class A). Its scoring reference can be historical or same-item-set; either way, separability need not yield power beyond clean-score variability. A *post-hoc certifier* recomputes a view from inputs it trusts (the evaluation contract recomputes R from $(\hat{y}, y)$) and needs trusted inputs rather than a stored value.

Proposition 4. (i) For any auditor whose decision is a function of $V_{\mathcal{I}}(s)$, $a \in B_{\mathcal{I}}$ implies $\mathrm{fire}(s_a) = \mathrm{fire}(s_0)$ pathwise under the same state and reference construction. This equality does not in general imply equality with a false-action rate estimated under a different clean-resampling or reference construction. (ii) A reference-anchored auditor detects every $a \notin B_{\mathcal{J}}$, deterministically.

Corollary 3 (which reference certifies which integrity). Let $a \in T_y$ with f fixed. (a) A trusted exact same-batch claim reference $R_0 = R(s_0)$ detects exactly every $R(s_a) \neq R_0$ and suffices for a conclusion-only claim. A trusted $M_0 = M(s_0)$ detects every violation of aggregate integrity and, by Proposition 3, every material conclusion change. (b) References that factor through M, $\mathrm{hist}(y)$ or R miss C1 relabelings; item identity needs level C, whose aligned y_0 detects every non-identity relabeling. Thus R_0, M_0 and item alignment protect conclusion, aggregate-plus-conclusion and identity claims, respectively. These are minimum levels within the declared reference lattice, not minima over every encoding or audit architecture.

Proposition 5 (union versus conformal family calibration). (a) If m sensors fire under the null with probabilities $p_j \leq \alpha$, then $\max_j p_j \leq \Pr_0[\bigcup_j E_j] \leq \min(1, \sum_j p_j) \leq \min(1, m\alpha)$, the bounds being attained by nested and disjoint events. (b) Let $v_1, \ldots, v_n$ be the sensor vectors of n calibration draws and v_{n+1} that of the audited batch; give every member j of the augmented set the leave-one-out family score $\tilde{U}_j = \max_s \#\{i \neq j : v_{s,i} < v_{s,j}\}/n$, let $p = (1 + \#\{i \leq n : \tilde{U}_i \geq \tilde{U}_{n+1}\})/(n+1)$, and fire iff $p \leq \alpha$ (ties against firing). At most $\lfloor \alpha(n+1) \rfloor$ members of any augmented set would fire if audited; hence, if the $n+1$ draws are exchangeable, $\Pr_0[p \leq \alpha] \leq \lfloor \alpha(n+1) \rfloor/(n+1) \leq \alpha$ with no continuity or tie assumption [30], [33], [34]; $10/201 = 0.0498$ for $n = 200$, $\alpha = 0.05$. For one sensor without ties the rule is the per-sensor rule of Gate N. The level is marginal over the joint draw of calibration set and audited batch; nothing is claimed when the premise fails. (c) An asymmetric construction that scored calibration draws against the other $n-1$ draws only, has no such guarantee: with three sensors and cyclic orderings over five vectors it fires with probability 0.60 at $\alpha = 0.2$ under exchangeability (supplement), and on the frozen draws it exceeds α under exchangeable re-splits where the conformal rule does not (Section VI-B).

Proposition 6 (no local authentication without an uncontrolled root). Let a local verifier be any map $\mathrm{acc}(V_{\mathcal{I}}(s), \rho)$ of the observed view and a reference bundle, with honest references $\rho_H(s) = V_{\mathcal{J}}(s)$. If it accepts every honest pair and the class can produce $(V_{\mathcal{I}}(s'), \rho_H(s'))$ for some honest $s' \not\sim_{\mathcal{I}} s_0$ (joint control), the substitution is accepted as an honest run of s': no function of that input establishes authenticity relative to s_0, and if $R(s') \neq R(s_0)$ a material change is served.

A bundle component $V_{\mathcal{K}}(s_0)$ the class cannot write rejects every substitution with $V_{\mathcal{K}}(s') \neq V_{\mathcal{K}}(s_0)$ (Proposition 4(ii)): authenticity requires a root outside the declared class. The result concerns authenticity, not consistency; by Proposition 3 a material substitution does change the joint view.

3.5 Counterexamples

Eight minimal constructions separate the events a robustness number conflates; the supplement tabulates them with the frozen expansion observations that realise each. C1 swaps the labels of two items with equal predictions: M, the histogram and R are invariant while item identity changes (808 of 3,600 label rows), an integrity violation with no conclusion impact, visible only item-wise against a trusted reference. C2–C4 separate marginal invariance, confusion-matrix invariance and conclusion invariance, C5–C6 do the same on the feature side, C7 shows that the clipped "harm" endpoint of the earlier evidence hid 1,534 conclusion changes (433 on the label path), and C8 is the empirical face of Proposition 3 (2,617 of 2,617 material label rows change M). Of the 3,600 label rows, a trusted aggregate reference of the same batch detects 2,792 (every material one, Corollary 3a) and the remaining 808 are C1 relabelings that only an item-aligned reference exposes (Corollary 3b). A ninth construction, the cluster-preserving drift of Section VI-D, is not a structural blind region but sensor insufficiency against an adaptive attacker, measured rather than proved.

3.6 The Quantum Branch as an Instance of the Lattice

Let $\mathcal{C}$ be the circuit representations (canonical OpenQASM 3 with bound parameters), $\Phi : \mathcal{C} \to \mathcal{K}$ the semantic map to the ideal fidelity kernel on a fixed probe set, $K_{\text{sem}} = \Phi(C)$, $\hat{K}$ and K_{obs} the estimated and observed kernels of Section III-A, $h(C)$ the provenance hash, $A(\cdot)$ the algebraic invariants and $f_K(\tilde{X})$ the decision computed from a kernel. Three intervention classes are kept apart: (a) circuit-side, overwriting C; (b) estimation variation, with C and K_{sem} fixed and $\hat{K}$ changing; (c) post-processing or kernel substitution, with C, K_{sem} and $\hat{K}$ fixed and K_{obs} changing. Each inclusion below names its class; none is claimed across classes.

Proposition 7 (quantum lattice). Assume h collision-free on the circuits under study. (i) [class (a)] $[C] = \Phi^{-1}(\Phi(C))$ is the semantic class of C; approved transpilation and common-unitary rewrites map C into $[C]$ without fixing $h(C)$, so provenance refines the semantic view, $B_h \subseteq B_{K_{\text{sem}}}$, strictly whenever the class contains an approved rewrite: separability in provenance is not harm, and the auditor needs the approved class, implemented as semantic equality of probe kernels against the trusted $K_{\text{sem},0} = \Phi(C_0)$ (level B). (ii) [class (c)] A and f_K coarsen the observed kernel, so $B_{K_{\text{obs}}} \subseteq B_A$ and $B_{K_{\text{obs}}} \subseteq B_f$; a PSD-preserving substitution K' with $A(K') = A(K_{\text{obs},0})$ lies in $B_A \setminus B_{K_{\text{obs}}}$ and, since $h(C)$ and the semantic probe of the unchanged circuit see nothing, is closed only by an anchored comparison of K_{obs} itself: the trusted semantic probe and the trusted observed-kernel reference are different anchors. A kernel change that crosses no decision boundary lies in $B_f \setminus B_{K_{\text{obs}}}$. (iii) [class (b)] For an honest finite-shot estimate, exact equality $\hat{K} = K_{\text{sem},0}$ has false-alarm probability $1 - \Pr[\hat{K} = K_{\text{sem},0} \mid \text{honest}]$, which depends on the discrete support of the estimator and may be large or equal to one but is not universally one (fidelity $1/2$ at two shots gives $1/2$). Exact equality is therefore not an appropriate acceptance criterion; $d(\hat{K}, K_{\text{sem},0})$ must be treated as a level-A statistic with a null from repeated honest estimation, to which Proposition 5(b) applies with one sensor when the repeated and audited estimates are exchangeable, and where no such null is calibrated no statistical integrity claim is made. The proposition adds no quantum mechanics: it names the two features the classical branches lack, an approved equivalence class coarser than provenance and an estimation step that turns an exact anchor into a statistical one. Hash chaining of the audit envelope is a reference-anchored comparison over the record itself; its root is not assumed uncontrolled, so Proposition 6 applies.

4 Adversary and Failure Model

Table 1 assigns every executed intervention to one of four classes with the roots it cannot write and the least separating evaluated regime (complete per-class record in the artifact). Evaluation-label classes stand for corruption of the ground-truth store or label join; feature-side classes for corruption or drift of the acquisition path feeding both branches; the cluster-preserving class for an attacker who knows the batch-level fingerprint; circuit and kernel classes for corruption of the circuit/parameter store, the compiler output or the post-estimation kernel; shot emulation for legitimate estimation uncertainty. Equal weighting is a stress profile, not a threat distribution; no prevalence is estimated.

The evidence trusts the local runtime and host, the training data and fitted model, the reference input and preprocessing declaration, the reference circuit and probe kernels, SHA-256 collision resistance, the item-level label reference only where $\mathcal{I}^{\star}_{XFY}$ is declared, and the code that builds and verifies the envelope. Compromise of the operating system, runtime or verifier; provider identity, forged job or calibration metadata and provider attestation; malicious schedulers, physical mapping and unapproved passes on hardware; device-calibrated noise, crosstalk, multi-tenant interference and QPU faults; side channels and circuit confidentiality; denial of service; and the operational Fleet Management System are outside this study.

5 Methodology

5.1 Data, Pipelines and Interventions

The frozen first gate uses a balanced binary subset of CICIDS2017 [58]: 3,000 rows, 77 numeric columns and 128/128 capped training/evaluation samples. The expansion adds eight fixed environments E1–E8 (supplement): five from CICIDS2017 (a 256/256 scale gate and four temporal source-target pairs), two from UNSW-NB15 [59] (a balanced ID subset and a temporal pair) and one from ToN-IoT [60] (a balanced ID subset), all at 128/128 unless stated and projected to dimensions 8, 10 and 12. CICIDS2017 therefore carries six of the nine environments including Gate 1; the environments are not a probability sample of deployments. Train-defined preprocessing is applied without evaluation labels: TruncatedSVD per branch, training-fitted standardization for the RBF support-vector classifier (SVC)

TABLE 1
Adversary and failure model of the executed interventions. Roots: assets the class cannot write. Least separating evidence in the frozen design: the evaluated regime that first separates the class under the conformal family rule; "exact" means an item-aligned trusted reference; (a)–(c) are the intervention classes of Proposition 7.

Mechanism and capability (boundary)	Class	Roots it cannot write	Least separating evidence in frozen design	Supported claim
Random label flip: rewrite r of y at random (evaluation/report)	Integrity corruption	X, P, C, K, f; item-aligned $y^\star$ if declared	exact only; $\mathcal{I}_{Y_m}/\mathcal{I}_{XFY}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}$ (Cor. 1a); undetected at budget by marginal or batch joint evidence at these rates
Prior-preserving label flip: pairwise swaps preserving class counts (evaluation/report)	Adaptive attacker	as above	exact only; $\mathcal{I}_{XFY}$ batch 0.00–0.01	invisible to $\mathcal{I}_X, \mathcal{I}_{XF}, \mathcal{I}_{Y_m}$ (Cor. 1); every material instance changes M (Prop. 3)
Confusion-preserving relabeling: swaps among equal predictions (realised, not executed)	Adaptive attacker	as above	item-aligned reference only	item-identity violation without aggregate or conclusion impact (C1; 808 rows)
Feature sign flip: negate p of $\tilde{X}$ entries (acquisition/preprocessing)	Integrity corruption	training data, fitted model, reference features	$\mathcal{I}_X$ 0.09–0.72; $\mathcal{I}_{XF}$ 0.13–0.84; exact 0.79–0.91	partially separable; strength-dependent residual
Mean shift, scaling drift: shift or scale all features (acquisition/preprocessing)	Fault robustness	as above	$\mathcal{I}_X$ 0.99–1.00 by cluster fingerprint	detected in this design by a fragile fingerprint
Feature dropout: impute p of entries (preprocessing)	Fault robustness	as above	batch 0.00–0.03; exact 0.49–0.76	separable in $\mathcal{I}_{XF}$ yet missed by the executed finite-batch level-A rule
Cluster-preserving mean shift and scaling: same mechanisms on the 32% of entries outside tight clusters (executed, Gate A)	Adaptive attacker	as above	$\mathcal{I}_X$ 0.01–0.56, $\mathcal{I}_{XF}$ 0.06–0.66 at matched strengths; exact for every material row	evades the fingerprint while keeping 83–91% of the conclusion changes; served by P2 in 28–39% of material rows
Circuit-parameter mutation; repetition change: alter C before execution (design; (a))	Adversarial attack	reference circuit hash, probe kernel $K_{\mathrm{sem},0}$	circuit hash and semantic probe 15/15	identified by anchored provenance and semantics; outputs miss subdecision changes
Benign transpilation; common-unitary rewrite: compiler pass ((a))	Approved equivalence	–	hash changes 15/15, K_{sem} unchanged	separability in $h(C)$ is not harm (Prop. 7(i))
Asymmetric, diagonal-eroded kernel edit: edit K_{obs} after estimation ((c))	Integrity corruption	reference kernel, algebraic invariants	algebra 15/15	detected by structural invariants of K_{obs}
PSD-preserving kernel substitution: valid-looking K_{obs} ((c))	Adaptive attacker	reference kernel	algebra 0/15; circuit hash 0/15; anchored K_{obs} comparison 15/15	sensor-level blindness closed only by a reference on the observed kernel (Prop. 7(ii))
Binomial shot emulation: finite-shot estimate (execution; (b))	Fault robustness	–	repeated estimation 15/15 in these cells	held for adjudication; exact equality is not the criterion (Prop. 7(iii)); not hardware evidence
Post-hoc envelope edit: rewrite stored contracts (record)	Adversarial attack	none assumed	hash chain	tamper-evident only; no identity or non-repudiation

and frozen angular scaling to $[0, 2\pi]$ for the fidelity-kernel branches (ZZ, PauliXYZ and a Z/PauliXZ-equivalent profile, one repetition). Gate 1 uses the Qiskit Machine Learning reference evaluator [61]; the expansion uses a tagged exact-statevector evaluator validated against it at 10^{-10} on unit matrices and by an end-to-end replay (supplement); neither is shot-noise or QPU evidence. The intervention suite has six mechanisms at strengths 0.02, 0.05 and 0.10: feature sign flip, feature-wise mean shift, scaling drift, feature dropout with median imputation, random evaluation-label flip and prior-preserving evaluation-label flip. Feature-only sensors are feature Jensen–Shannon divergence (JSD), maximum mean discrepancy (MMD) and the Kolmogorov–Smirnov (KS) rejection rate against the clean version of the same item set; prediction, label-marginal and joint-outcome evidence likewise compare aggregate score, class and confusion-profile distributions with their clean same-item-set values. None uses item pairing. The runner protects these clean arrays as an experimental oracle; the policy API receives only sensor and family flags, but producing those flags still requires the upstream oracle. This is statistical class A, not evidence that a deployed system authenticates the reference. Under $\mathcal{I}^\star_{XFY}$, confusion-profile deltas against a trusted aggregate are exact class B invariants, while prediction disagreement and label mismatch use trusted item alignment (class C). The complete sensor-to-reference audit is in the supplement.

The duplicate, clustered and near-constant structure on which this benchmark fingerprint depends is consistent with documented CICIDS2017 data-quality limitations [62]; the frozen staging is described rather than re-engineered here.

5.2 Experimental Units and Uncertainty

Gate 1 crosses five split seeds with four nested model seeds; its 7,980 raw rows contain 3,420 exact repeated SVC rows, leaving 4,560 unique observations. The expansion crosses the same five split seeds with two model seeds; all 360 job pairs completed and 13,680 raw rows contain 2,280 verified repeats, leaving 11,400 unique observations. Coverage endpoints are exact counts in the prespecified design; for the secondary model-profile endpoint (supplement) the five split seeds are the inferential units with two-sided 95% Student-t intervals (four degrees of freedom), describing within-environment resampling only.

5.3 Null Calibration and Decision-Level Calibration

The prespecified null-calibration gate calibrates the ten distributional sensors per (environment, dimension, model) cell at a nominal per-sensor $\alpha = 0.05$: the threshold is the 191st of 200 clean calibration draws from one half of the held-out evaluation pool, and false alarms are measured on 200 draws from the disjoint other half. Gate F calibrates each regime as a family with Proposition 5(b)'s conformal rule at $\alpha = 0.05$ per decision, using only the calibration draws. The same draws also evaluate the union and asymmetric comparison rules (Proposition 5(c)). The asymmetric rule's pooled excess over nominal is decomposed into rule bias (its rate minus the conformal rate on identical draws) and design effect (the conformal rate minus nominal). Both rules are also evaluated under 30 exchangeable re-splits of the 400 pooled draws per cell. These re-splits diagnose the rule construction without changing the nonexchangeable primary design. Because a run's 20 draws overlap within one pool half, false-alarm inference uses (environment, split-seed) clusters. Pooled rates are descriptive; intervals are five-cluster t intervals per environment. E1 (256-row draws from 322-row halves) remains in the preregistered primary aggregate and is also reported separately.

5.4 Decision Layer and Offline End-to-End Evaluation

Gate D composes sensors, the conformal-rule or union decision, the information regime, the trusted-reference status, a materiality threshold and the declared residual blind region into one `allow/hold/block` decision. `block` is reserved for violations of an exact invariant against a trusted reference; `hold` answers statistical evidence or, under P3, missing coverage; the decision composes with the four frozen contracts by the maximum in `allow` < `hold` < `block`. Four policies are fixed and named by what they are: P0 serve-always, the baseline; P1, the uncalibrated union rule of Gate N (risk-tolerant); P2, the conformal family rule, a *conformal-rule risk-tolerant* policy, nominally calibrated under exchangeability, that may serve under an explicitly declared residual blind region; P3, the *coverage-complete abstaining* variant, which holds whenever a mandatory protected boundary (feature, prediction, label) has neither exact nor declared statistical coverage under the regime. P3 fails closed on missing coverage, not on insufficient power: a boundary covered by a low-power statistical sensor is served as under P2, so P3 guarantees no minimum detection power. The prototype's contracts fail closed on their invariants; P2 does not and is never called fail-closed. Five regimes are evaluated: $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$, $\mathcal{I}_{XFY}$ (class-A statistical aggregate comparisons) and $\mathcal{I}^{\star}_{XFY}$ (class-B aggregate and class-C item-aligned exact invariants; feature coverage remains statistical). The policy consumes union/family/exact flags, not clean arrays; the upstream scorer has the reference dependency described above. The observation set is frozen: 12,000 clean evaluation draws, 1,200 near-null synthetic controls applied in place (Gaussian noise $\sigma = 0.001$, scaling $\alpha = 0.001$; not operational benign traffic), and the 10,800 intervened observations of the eight environments, 7,008 of which change the reported balanced accuracy (structural-sensitivity endpoint; $\tau = 0.02$ and 0.05 are larger-effect sensitivities, not operational risk thresholds). The primary endpoint is a materially altered audit result served: a material observation decided `allow`. The CSV retains `unsafe_allow` as a compatibility identifier; *unsafe* here means serving a materially corrupted audit/report conclusion, not allowing a malicious network event and not necessarily changing the classifier's live prediction. The interruption endpoints are the clean false-action rate (clean draws not allowed; for $\mathcal{I}^{\star}_{XFY}$ the denominator is the 1,200 exact-zero rows) and the interruption rate on the near-null controls (the same 1,200 rows for every regime), decomposed into statistical holds and exact-reference blocks. The evaluation is offline, on frozen outputs, not a deployed runtime service; twelve consistency checks fail closed, and every table and figure is generated from the manifested outputs. The 24,000 policy rows derived from the frozen intervention grid are not independent episodes; inference resides at environment/split clusters rather than at row level.

5.5 Adversarial Gate and Geometry Sensitivity

Gate A executes the preregistered cluster-preserving attacker. The standardized projected features hold tight clusters of rows (in every environment some feature keeps 69–87% of the rows within 0.01 standard deviations) that a fresh clean batch reproduces and any in-place displacement smears, which is why mean shift and scaling drift were detected in every cell. The attacker holds the batch, knows the sensor definitions and the fingerprint, and applies the executed mechanisms unchanged to every unclustered entry (an entry is clustered when at least $\lceil 0.05n \rceil$ rows, never fewer than two, lie within 0.01 batch standard deviations of it); clustered entries and labels are untouched. Strengths 0.02, 0.05, 0.10 match the frozen suite and 0.25, 0.50 probe whether materiality can be bought on the unclustered entries; the executed mechanisms run in the same jobs as matched controls and must reproduce the frozen expansion exactly. The gate runs in the eight environments with the frozen maps, caps and seeds (240 exact-statevector jobs) and is scored with the calibration draws, rules and policy layer of Gates F and D; window, minimum mass and strength grid were fixed before execution.

The geometry-aligned sensitivity leaves frozen evidence intact and replays the same 240 configurations. For fixed E, calibration is $s(E, C_k)$ and aligned clean versus intervention scoring is $s(E, B)$ versus $s(E, T(B))$, with fresh B fixed before results. Exact identity $s(B, B)$ is separate from clean resampling. There is one row per frozen model cell and intervention; no label attack, training or selection is rerun (supplement).

5.6 Quantum Gate and Executable Prototype

A separate simulator gate isolates the quantum boundary with CICIDS, 64/64 samples, five split seeds, dimensions 4, 6 and 8 and 11 conditions (165 cells): clean, benign level-1 transpilation and a common post-feature-map rewrite as controls; data-dependent RZ mutations at 0.02 and 0.10; a feature-map repetition change; asymmetric, diagonal-eroded and PSD-preserving kernel edits; and binomial fidelity-estimation emulators at 256 and 1,024 shots. Each cell records canonical OpenQASM 3 provenance, kernel hashes, the anchored comparison against the trusted reference kernel, algebraic checks, repeated-estimation discrepancy and prediction changes. The prototype instantiates four ordered contracts (input/preprocessing, circuit/kernel, execution/result, evaluation/report) in an SHA-256 chain over one compact CICIDS cell and six prespecified scenarios; it evaluates composition and decision logic, not throughput. It is a methodological V&V and integrity contribution within Deusto's ATHENA-AEGIS work on hybrid workflows, not evidence of full ATHENA validation, deployment, provider attestation, confidentiality, QPU protection or Fleet Management.

6 Results

6.1 Exact Label-Path Blind Regions

The zero-response cells are validation checks of the structural prediction, not findings: Corollary 1 says that a sensor whose input does not change cannot respond, and the checks confirm that the implementation realises the declared views. All 3,600 expansion and 1,440 Gate-1 evaluation-label observations (of 11,400 and 4,560 unique observations) have zero prediction disagreement and exactly unchanged feature JSD, MMD, KS rejection and score JSD; all 1,800 and 720 prior-preserving observations also have zero prior shift and label JSD; the verifier recomputes every count. The informative quantity is how often these blind regions coincide with material conclusion changes and, in Section VI-C, with serving decisions. The signed conclusion change is not one-sided: in the expansion 2,184 label observations lower the reported balanced accuracy, 983 leave it unchanged and 433 raise it (Gate 1: 1,276, 105, 59). Every one of the 2,617 (Gate 1: 1,335) label observations with a changed conclusion has non-zero item-aligned confusion evidence, and every one of the 2,184 with a lowered conclusion, the count used by the one-sided diagnostic, is among them.

6.2 Decision-Level Calibration

Over the 12,000 evaluation draws the per-sensor false-alarm rates lie between 0.029 and 0.069, but the batch-level rule “fire if any sensor fires” is not calibrated at the level of the decision: its rates are 0.125, 0.203, 0.048 and 0.259 for $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{Y_m}$ and $\mathcal{I}_{XFY}$ (three, six, two perfectly dependent and ten sensors). The conformal family rule brings them to 0.056, 0.058, 0.048 and 0.053 (Fig. 1(a)), observed rates in a design that violates the exchangeability premise. For comparison, the asymmetric construction gives 0.061, 0.073, 0.048 and 0.079 on the same draws. Its excess over 0.05 decomposes into rule bias +0.005, +0.015, +0.000 and +0.025, and design effect +0.006, +0.008, -0.003 and +0.003, and under exchangeable re-splits of the pooled draws, where any excess is attributable to the construction, it fires at 0.052, 0.058, 0.040 and 0.059 while the conformal rule stays at 0.044, 0.042, 0.040 and 0.036 against its level 0.0498. What remains with the conformal rule is the design effect, concentrated in E1, whose 200 draws per cell carry the information of about one independent batch per pool half (0.125, 0.115, 0.123 in $\mathcal{I}_X$, $\mathcal{I}_{XF}$, $\mathcal{I}_{XFY}$); without E1 the conformal rule sits at 0.048, 0.052, 0.048 and 0.045; E1 stays in the primary aggregate. A split construction (0.049, 0.039, 0.067, 0.034; supplement) is not adopted because it splits the draws between ranking and calibration, reducing the calibration sample and coarsening p-value resolution to $1/101$.

As a consistency check of Proposition 4(i), the executed rules confirm the pathwise structural prediction: on all 3,600 label observations the per-sensor and the conformal $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ rules, and on all 1,800 prior-preserving observations the $\mathcal{I}_{Y_m}$ rules, fire zero times, as they must when their input is unchanged. The informative result is on the other side: the batch-level $\mathcal{I}_{XFY}$ rules detect 43 (union) and 11 (conformal) of the 2,617 material label observations, 0.016 and 0.004, although every one of them is separable (Proposition 3, C8). These 11–43 detections among 2,617 material rows are finite-batch statistical-power results, not structural blind regions: in this design a 2–10% relabeling of a 128-row batch can shift the confusion profile by less than its variability across fresh clean batches. A trusted aggregate reference of the same batch detects all 2,617 exactly (Corollary 3a) and 2,792 of the 3,600 label rows in total; the 808 relabelings that preserve every aggregate (C1) are exposed only by the item-aligned label reference (Corollary 3b). Feature-side coverage is mechanism-dependent: mean shift and scaling drift are detected in 0.99–1.00 of $\mathcal{I}_X$ cells, 0.96–1.00 of $\mathcal{I}_{XF}$ cells and 0.95–1.00 of $\mathcal{I}_{XFY}$ cells because the standardized projected features hold tight clusters that an in-place perturbation smears and a fresh batch reproduces, a legitimate but fragile fingerprint that also fires on the near-null controls (40–64% with the conformal rule); feature dropout with imputation is nearly invisible at the batch level (0–3%) yet changes predictions in 49–76% of cells. Within this fixed intervention grid, the conformal rule reduces detection only slightly: containment in $\mathcal{I}_{XF}$ is 0.40 under the union and 0.38 under the conformal rule at the primary threshold, 0.70 versus 0.66 at $\tau = 0.05$.

6.3 Offline End-to-End Decisions

Table 2 and Fig. 1(b,c) report the offline end-to-end evaluation. All aggregate served fractions summarize the prespecified equal-weight intervention grid; they are not deployment prevalence or expected operational unsafe rates. Under P0 every one of the 7,008 material observations is served. Under the batch-level regimes the conformal-rule policy P2 serves 4,496 ($\mathcal{I}_X$), 4,365 ($\mathcal{I}_{XF}$) and 4,390 ($\mathcal{I}_{XFY}$) of them with clean false-action rates 0.056, 0.058 and 0.053, whereas the uncalibrated union serves 4,429, 4,231 and 4,180 at 0.125, 0.203 and 0.259: the union buys 0.40 versus 0.37 containment in $\mathcal{I}_{XFY}$ at more than four times the clean false-action rate and 0.76 versus 0.49 interruption of the near-null controls. The asymmetric comparison rule served 4,494, 4,327 and 4,322; the adopted rule moves the count by less than 1% of the material rows. The served material corruptions are dominated by structure, not by thresholds: all 2,617 material label observations are served under $\mathcal{I}_X$ and $\mathcal{I}_{XF}$ (residual blind region), 2,606 of them under batch-level $\mathcal{I}_{XFY}$ (missed by the executed finite-batch rule), and 1,057 of 1,079 material dropout observations under $\mathcal{I}_{XF}$. The $\mathcal{I}_{Y_m}$ view provides zero containment of material interventions in this suite; 5,450 material cases remain in its residual blind region at $\tau \to 0^+$, while all 7,008 material cases are served once both blind and statistically missed cases are counted.

Containment 1.00 follows constructively from Corollary 3 under the declared trusted references; the experiment quantifies interruption within the prespecified near-null stress controls. Consistently, the trusted-reference regime serves

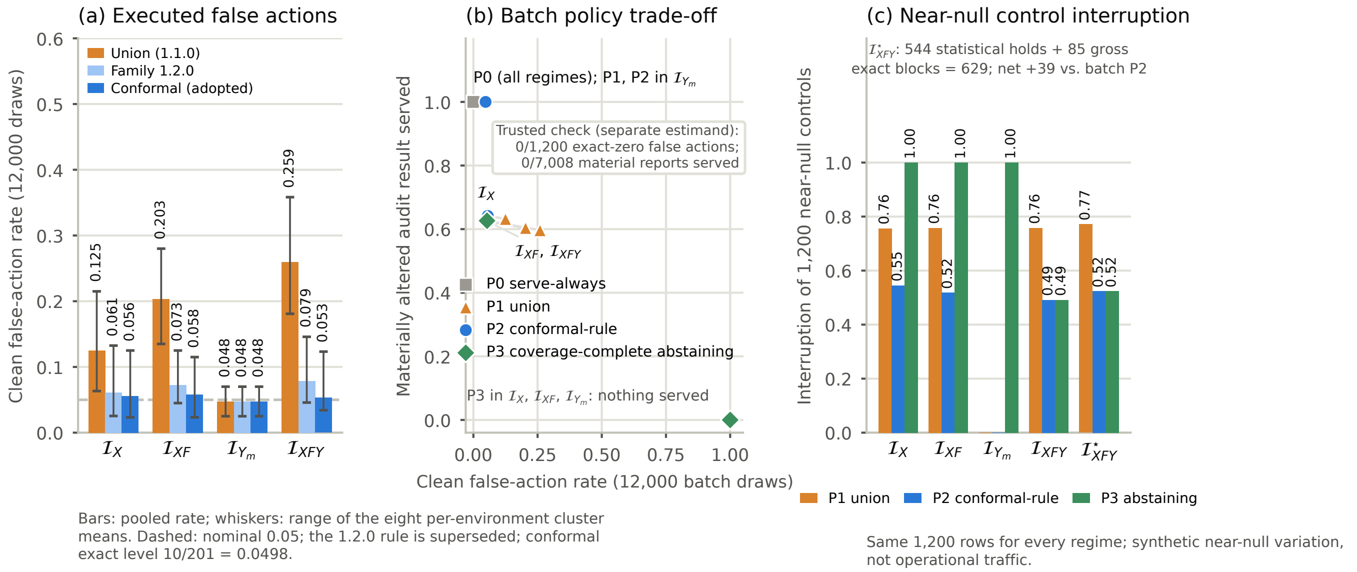

Fig. 1. Gates F and D (prespecified frozen evidence). (a) Decision-level false-alarm rate on 12,000 disjoint clean draws of the union rule, of the asymmetric comparison rule and of the conformal family rule per regime; whiskers span the eight per-environment cluster means. (b) Rate of materially altered audit results served against clean false-action rate for each batch policy and regime at the structural-sensitivity endpoint; the batch FPR denominator is 12,000 draws. The trusted invariant check is reported separately because its denominator is 1,200 exact-zero rows and it is a different estimand. The coverage-complete abstaining policy P3 serves nothing where the label boundary has no coverage. (c) Interruption of the same 1,200 near-null synthetic controls by each policy and regime; the trusted regime's origin in (b) interrupts 629 controls here, 544 statistical holds and 85 gross exact-reference blocks.

none of the 7,008 material observations with zero clean false actions on its 1,200 exact-zero rows and blocks every non-identity relabeling and every prediction change through its item-aligned references. Its stress-control interruption has two parts. Of the 1,200 near-null synthetic controls the trusted conformal-rule policy interrupts 629: 544 are statistical holds by its batch component, and 85 are gross exact-reference blocks of controls whose predictions changed under a 0.001 perturbation. Of those exact blocks, 46 already belong to the 590 interruptions produced by batch $\mathcal{I}_{XFY}$/P2. The net increment is therefore 39/1,200, or 3.25 percentage points, not the gross 85/1,200. The controls are synthetic near-null variation, not operational traffic, so neither number is a production false-positive rate, and the two denominators of Table 2 are reported side by side. The coverage-complete abstaining policy P3 makes information-set conditionality literal: in $\mathcal{I}_X$, $\mathcal{I}_{XF}$ and $\mathcal{I}_{Y_m}$ the label boundary has no coverage, so nothing is served and every clean and near-null row is interrupted; in $\mathcal{I}_{XFY}$ and $\mathcal{I}^\star_{XFY}$ every boundary is covered and P3 coincides with P2, including P2's 4,390 materially altered audit results served in $\mathcal{I}_{XFY}$: abstention on missing coverage is not a guarantee of power, and the choice between P2 and P3 is a declared risk decision, not a detector property. The composition with the six frozen contract envelopes reproduces their actions (supplement).

6.4 Adaptive Cluster-Preserving Attacker

The frozen Gate A stress test used clean-resample calibration but scored attacked batches as $s(E, T(E))$. Its matched response falls from 0.96–1.00 to 0.01–0.66 while retaining 83–91% materiality; this is design-specific and against the declared cluster-sensitive fingerprint, not evidence of general evasion of drift detectors. Dataset cluster structure contributes to the response, and generalization beyond this fingerprint and these staged datasets remains open.

The aligned sensitivity confirms the qualitative ordering. Exact paired identity is 0, distinct from the clean-resample false-action rate (family rates 0.056–0.058). Mean shift and scaling remain almost fully responsive; dropout is 0.068–0.320 and near-null response spans 0.568–0.822. Adaptive response is lower in 25–40 of 40 environment/split cells, never higher, with materiality of 0.262–0.667. Full strength/regime results are in the supplement. Same-item scoring amplified part of the original contrast, but aligned Gate A still shows

TABLE 2
Offline decisions on frozen observations at the structural-sensitivity endpoint $|\Delta_R| > 0$ (7,008 materially altered audit results among 10,800 interventions). P1: union; P2: conformal, risk-tolerant; P3: coverage-complete abstaining. FPR is estimated on 12,000 disjoint clean draws for batch regimes; the starred row instead reports an exact-zero invariant check on 1,200 rows and is not the same estimand. Benign is interruption of the same 1,200 near-null synthetic controls. Served: undetected materially altered audit results served, not malicious network events. Blind: structurally indistinguishable material rows.

Regime	Policy	FPR/check	Benign	Served	Contain.	Blind
$\mathcal{I}_X$	P1 union	0.125	0.76	4,429	0.37	2,617
$\mathcal{I}_X$	P2 conf.	0.056	0.55	4,496	0.36	2,617
$\mathcal{I}_X$	P3 abst.	1.000	1.00	0	1.00	2,617
$\mathcal{I}_{XF}$	P1 union	0.203	0.76	4,231	0.40	2,617
$\mathcal{I}_{XF}$	P2 conf.	0.058	0.52	4,365	0.38	2,617
$\mathcal{I}_{XF}$	P3 abst.	1.000	1.00	0	1.00	2,617
$\mathcal{I}_{Y_m}$	P1 union	0.048	0.00	7,008	0.00	5,450
$\mathcal{I}_{Y_m}$	P2 conf.	0.048	0.00	7,008	0.00	5,450
$\mathcal{I}_{Y_m}$	P3 abst.	1.000	1.00	0	1.00	5,450
$\mathcal{I}_{XFY}$	P1 union	0.259	0.76	4,180	0.40	0
$\mathcal{I}_{XFY}$	P2 conf.	0.053	0.49	4,390	0.37	0
$\mathcal{I}^\star_{XFY}$	P2/P3 (exact)	0.000	0.52	0	1.00	0

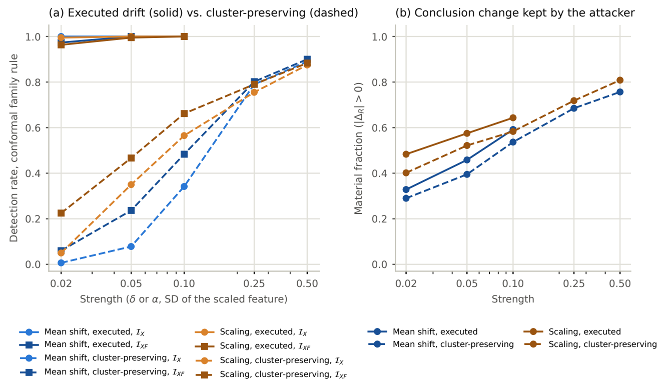

Fig. 2. Original frozen Gate A. (a) Detection under the conformal family rule of the executed drift mechanisms (solid) and of their cluster-preserving variants (dashed) in $\mathcal{I}_X$ and $\mathcal{I}_{XF}$, 600 observations per point. (b) Material fraction of the same rows: the attacker evades the batch-level sensors at matched strengths while keeping most of the conclusion changes.

that calibration controls clean-resample false actions without guaranteeing attack coverage.

6.5 Quantum-Workflow Coverage

All 165 design cells (11 conditions × five splits × three dimensions) and all nine acceptance checks pass. In this ideal-statevector gate one anchored comparison against $\Phi(C_0)$ is a semantic probe on circuit-side rows and an observed-kernel reference on post-processing rows; finite-shot estimation separates those anchors. Approved rewrites change provenance but preserve semantics (class (a)); PSD-preserving substitution escapes algebra and circuit provenance but not the anchored K_{obs} comparison (class (c)); both shot emulators yield non-zero repeated-estimation discrepancy in their 30 cells without establishing a universal rate (class (b)). Full per-condition responses, contract decisions and the secondary model profile are in the supplement. This is a bounded simulator instantiation of Proposition 7, not QPU/provider validation.

7 Integrity-Audit Contract

Each contract names its boundary, intervention class, view/reference, premise, residual region and response. For labels, R_0, M_0 and item alignment bind conclusion, aggregate and identity claims. The quantum contract combines provenance, approved equivalence, anchored kernels, algebra and repeated estimation. Exact-invariant violations block; missing mandatory evidence holds; P2 may serve within a residual region, while P3 promises no minimum power.

8 Discussion and Limitations

8.1 Security Interpretation

A metric can change while predictor output is unchanged: input robustness cannot repair corrupted evaluation labels. Zero response establishes blindness only when the evidence could not distinguish the states; otherwise power depends on the null. The 28–39% served fractions summarize the prespecified equal-weight intervention grid, not deployment prevalence. Gate A is a stress test of one cluster-sensitive fingerprint; its aligned sensitivity preserves the control/adaptive ordering while exposing a geometry effect.

The claim-relative minimum is within the declared reference lattice, where granularity means bound information, not digest length. A Merkle root or position-binding vector commitment can compactly bind level-C item identity [63], [64]. Level B instead lets aggregate- or conclusion-only audits verify M_0 or R_0 without retaining or exposing the complete item-wise label store, which can reduce disclosure and support separation of duties over sensitive labels. This is evidence minimization, not a new cryptographic primitive or legal guarantee. An external root remains necessary (Proposition 6); this is neither a deployed protocol nor a minimum over every possible audit architecture.

Within the prespecified near-null stress controls, 85 exact-reference blocks overlap 46 batch $\mathcal{I}_{XFY}$/P2 interruptions, so the incremental interruption is 39/1,200 (3.25 percentage points). This is not an operational traffic false-positive rate.

8.2 Validity Boundaries

The eight fixed settings are not sampled deployments, and balanced staging does not reproduce prevalence. The quantum branch uses ideal-statevector and finite-shot emulation, not QPU/provider validation. The secondary SVC/QSVC profile does not isolate kernel geometry. Structural blind regions are independent of the number of rows: a view-preserving intervention remains invisible as n grows. For interventions that do change an observed statistic, however, power can increase with batch size as sampling variability often decreases, approximately as $n^{-1/2}$ for proportion-like statistics under regular conditions. The 11–43 detections among 2,617 material label rows are therefore outcomes of the executed fixed-size design, not a universal bound on batch-monitoring power. Batch-size scaling is an important external-validity question and is not estimated by the present fixed-size experiment.

The conformal level requires exchangeability; overlapping draws, different pool rows and E1 violate it, so executed rates are descriptive. Gate A fixes one fingerprint and two mechanisms; generalization beyond its dataset structure remains open. The algebraic quantum check concerns the raw estimated/observed kernel before downstream PSD repair; a repaired-kernel consumer defines a different, unevaluated observable path.

The trusted item-aligned reference is an assumption under Proposition 6. P3 abstains on missing coverage, not low power. All policies are offline; runtime non-stationarity, recovery, QPU execution and service enforcement remain open.

9 Conclusion

Hybrid workflow integrity is evidence-relative. Interventions can be structurally invisible, statistically missed in a finite batch, or adaptive to the monitored fingerprint. The conformal guarantee remains conditional on exchangeability and the quantum instantiation simulator-bounded. Within the declared lattice, authenticated R_0, M_0 and item alignment certify conclusion, aggregate-plus-conclusion and item-identity claims, respectively. Assurance must state its view, trust root, premise, residual region and response.

Data, Code, and Ethics Statement

The artifact contains code, tests, frozen derived evidence and nine manifests; raw benchmarks are not redistributed. Sources, hashes and staging are documented. Zenodo version DOI: 10.5281/zenodo.22694063; concept DOI: 10.5281/zenodo.22550852. Code is Apache-2.0 and derived evidence/documentation CC BY 4.0. No new human-participant or personal-data collection was performed.

Acknowledgment

OpenAI Codex and Anthropic Claude Code assisted with draft/language review, code generation/inspection, literature-search support and artifact preparation across the manuscript and supplement. Human authors independently checked all claims, proofs, references, experiments, results and generated text using the stated tests, verifiers and primary sources, and assume full responsibility. Neither system is an author or proof authority; detailed disclosure accompanies the submission.

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